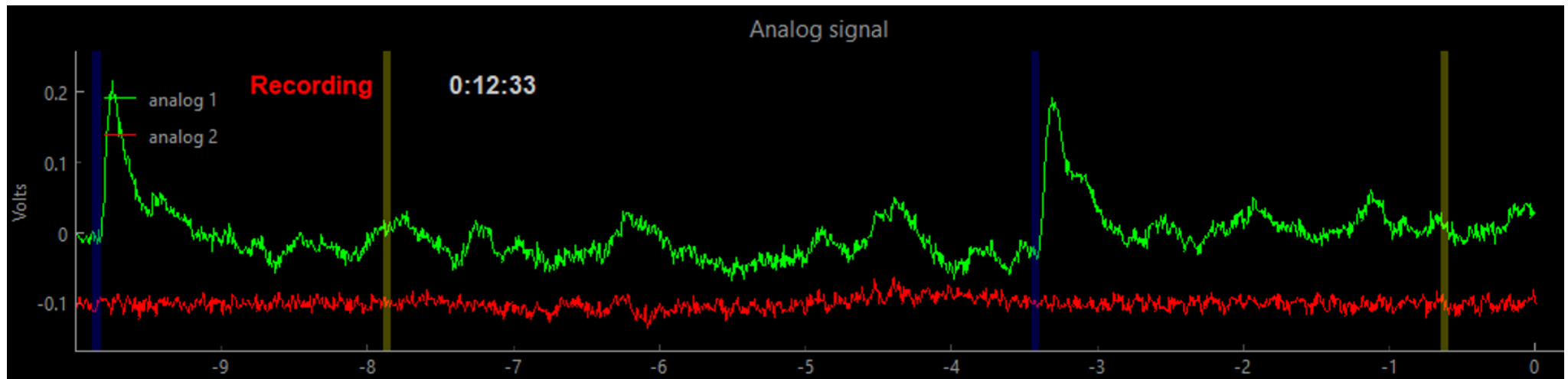
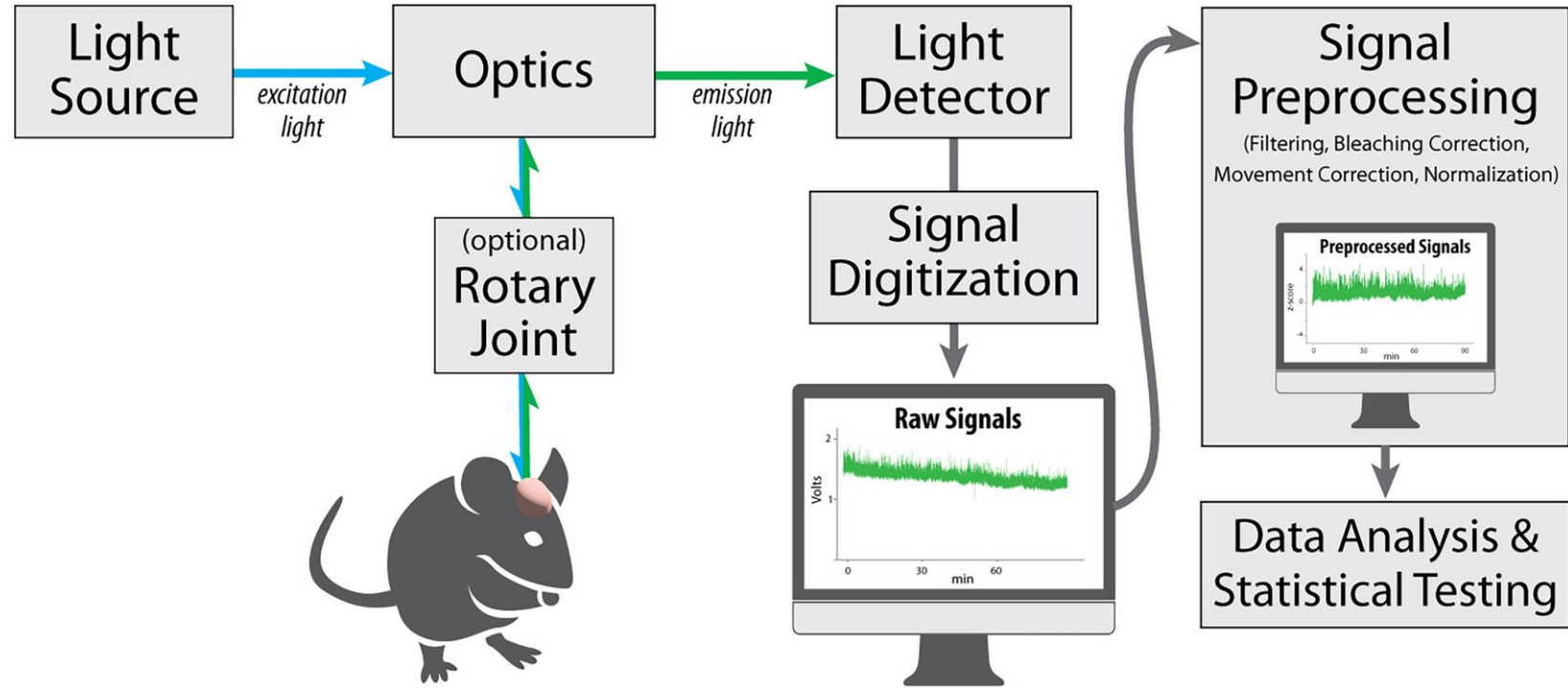


Fiber photometry

Data acquisition, preprocessing and analysis

Thomas Akam
Coimbra
6/2/2026

Fiber photometry



Fiber photometry

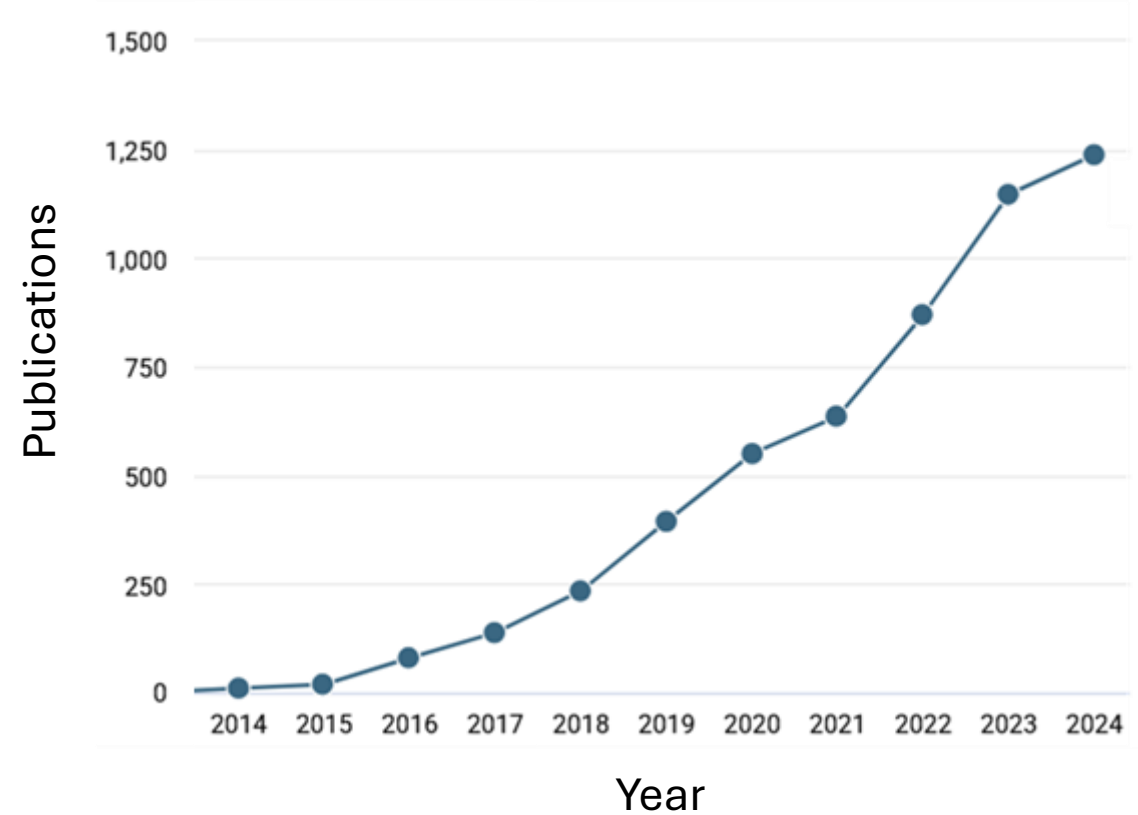
Strengths:

- Molecular / cellular specificity
- Rapidly expanding library of fluorescent sensors
- Access deep brain structures
- Wide range of timescales
- Multiplex multiple signals
- Simple and fairly cheap to implement

Limitations:

- Bulk signal
- Risk of artifacts

Publications including term “fiber photometry”

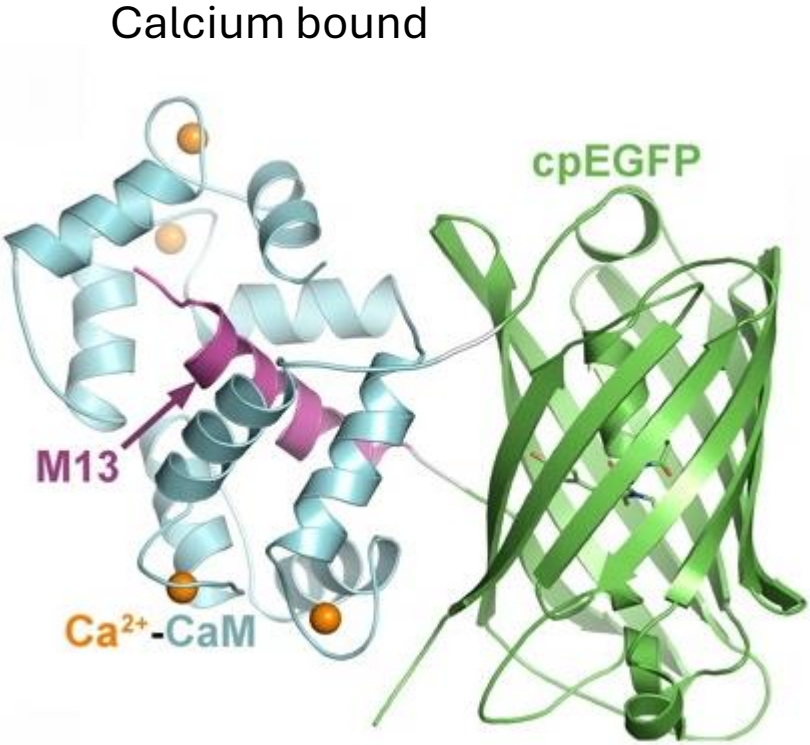
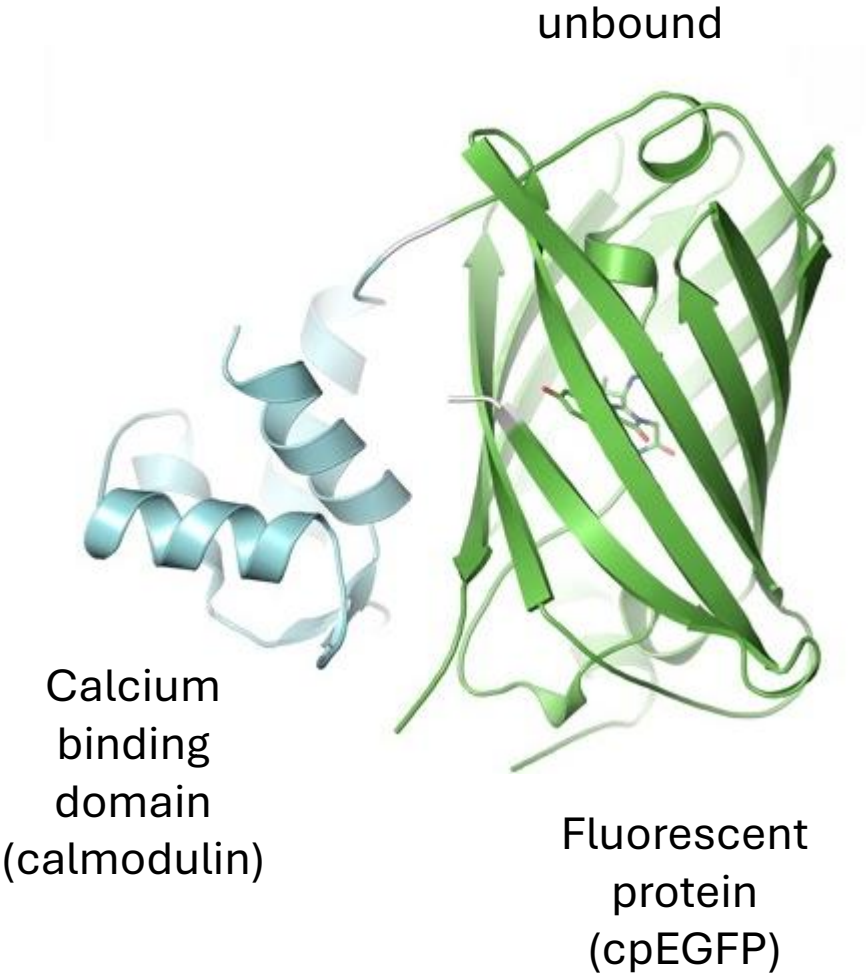


Talk outline

- Photometry data acquisition
- Photometry data preprocessing
- Photometry data analysis

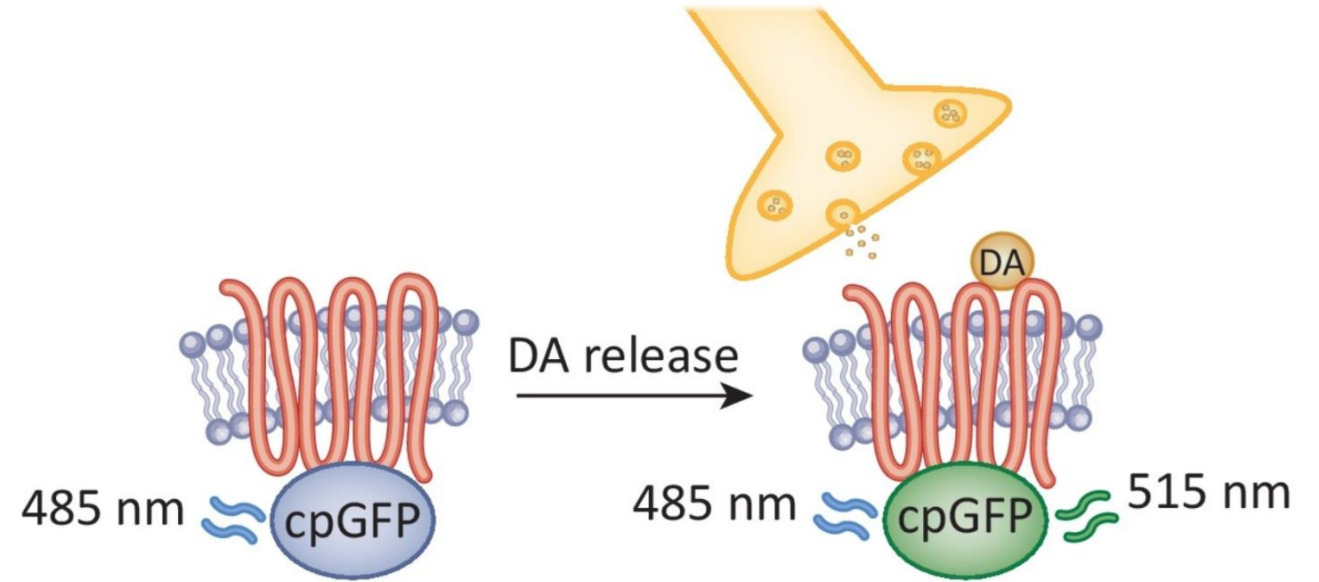
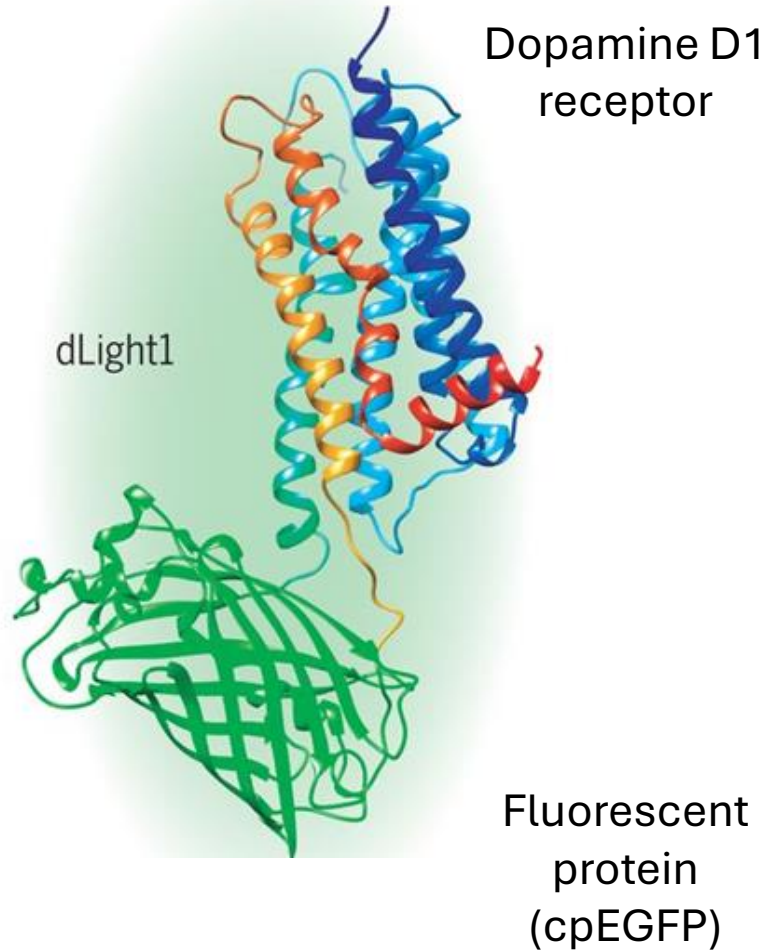
Genetically encoded fluorescent sensors

GcAMP

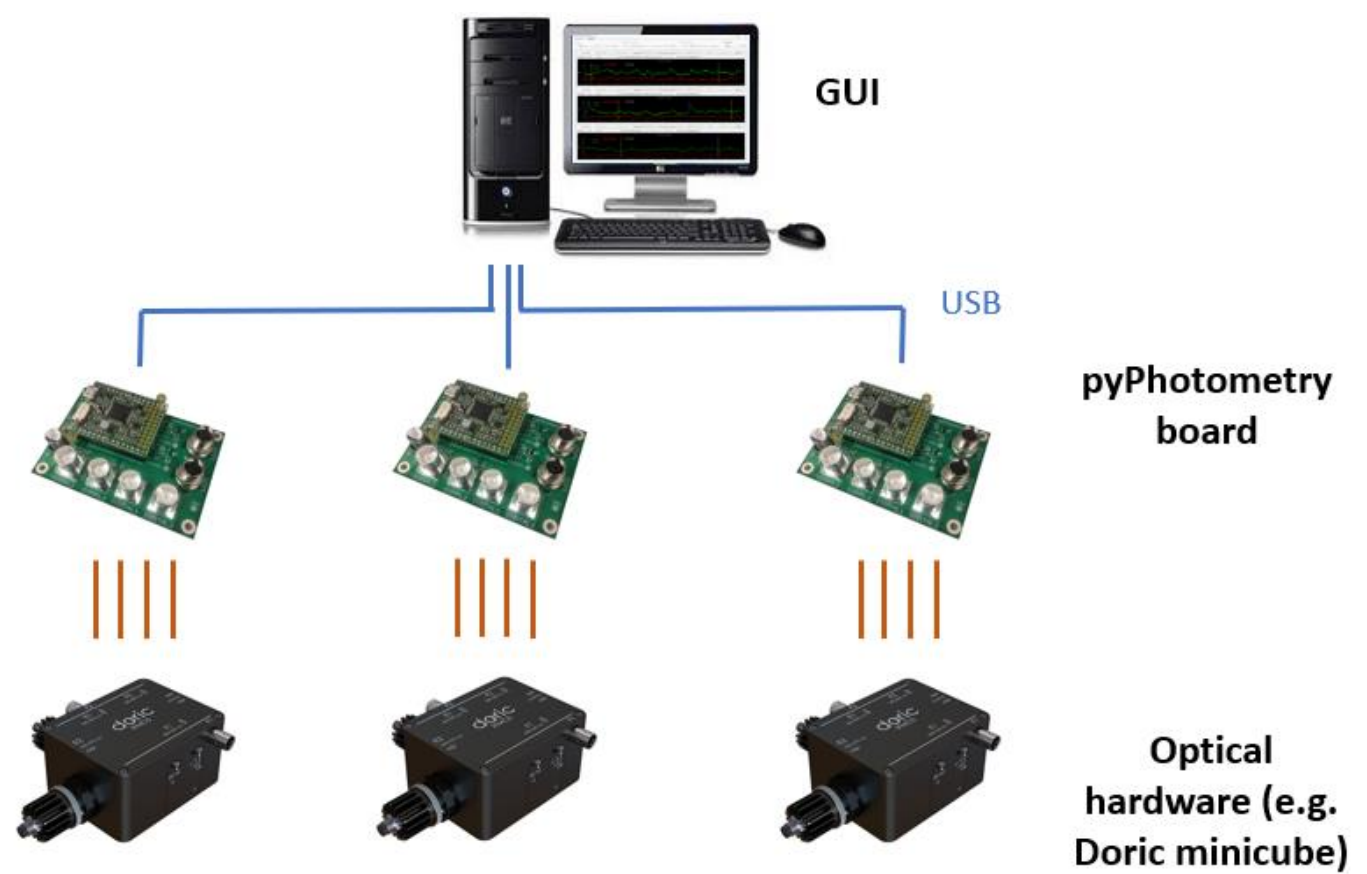


Genetically encoded fluorescent sensors

dLight



pyPhotometry overview



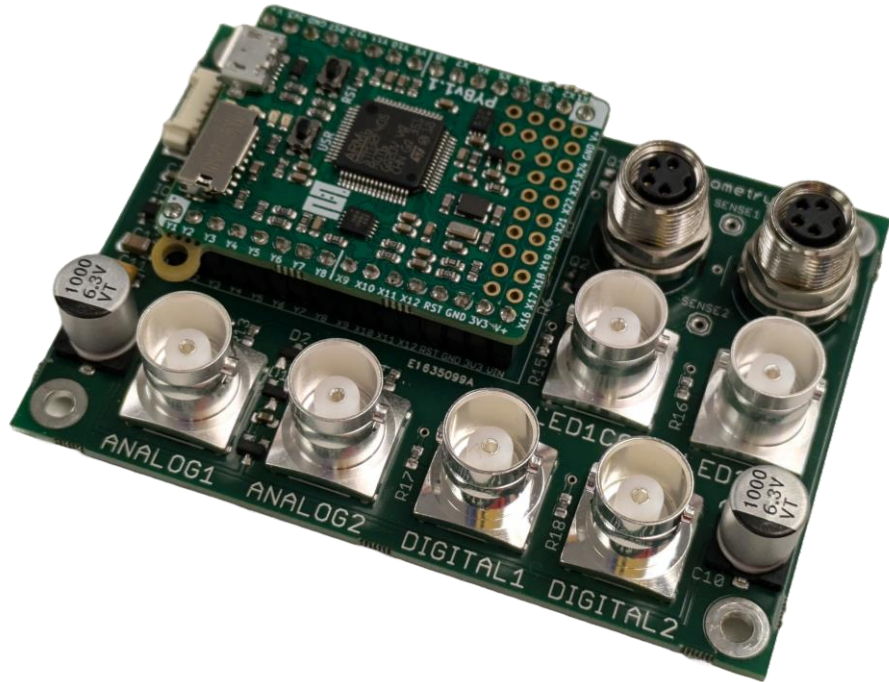
Aims:

- Reliable, user-friendly, low-cost, open source acquisition hardware and software

Components:

- pyPhotometry board
- Graphical user interface
- Commercial optics and photoreciever (we use Doric mini-cube)

pyPhotometry board v2.0



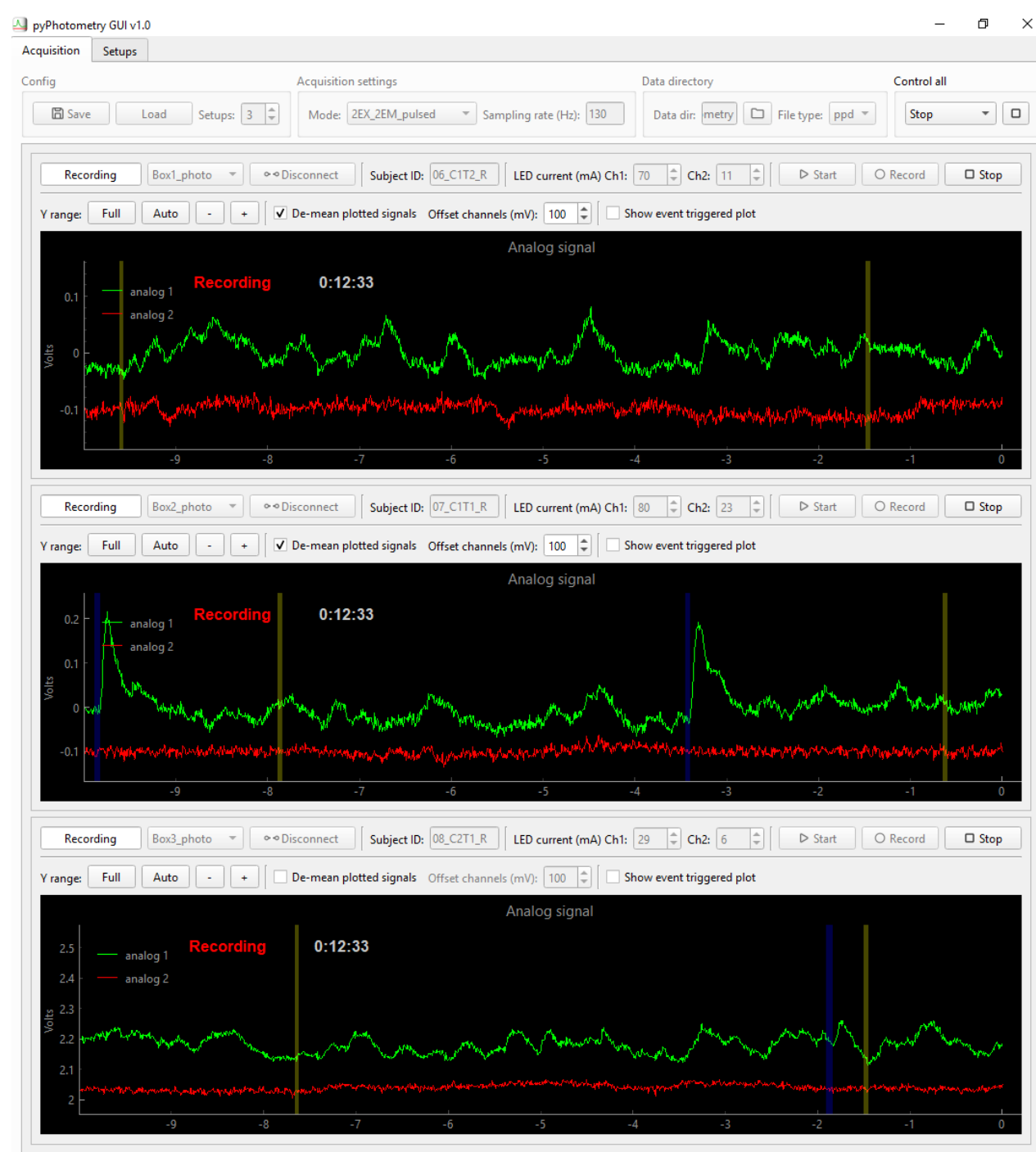
- Max LED driver current increased to 300mA
- LED control outputs (0-3.3V analog voltage)
- Expansion connector for access to microcontroller pins.



Integration with Doric Gen 3 minicube

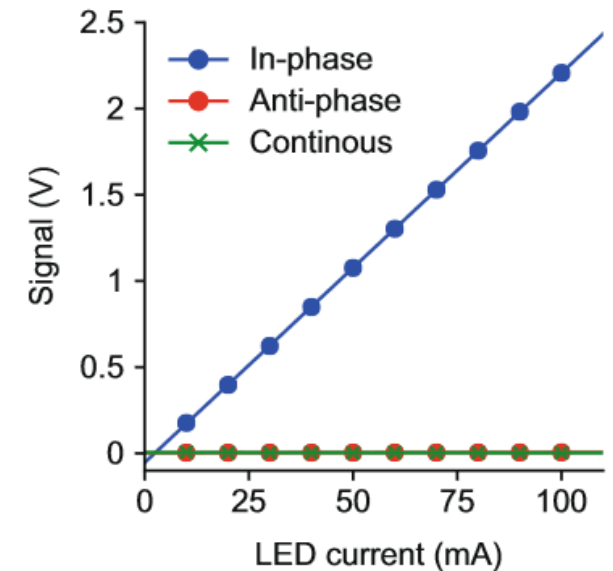
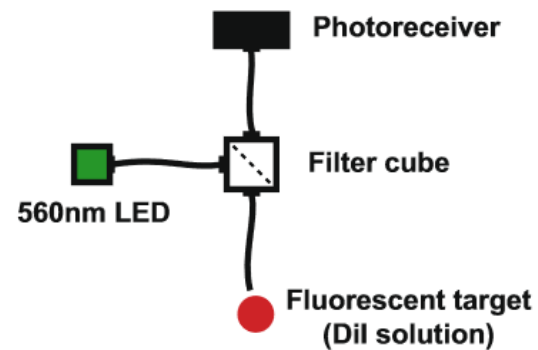
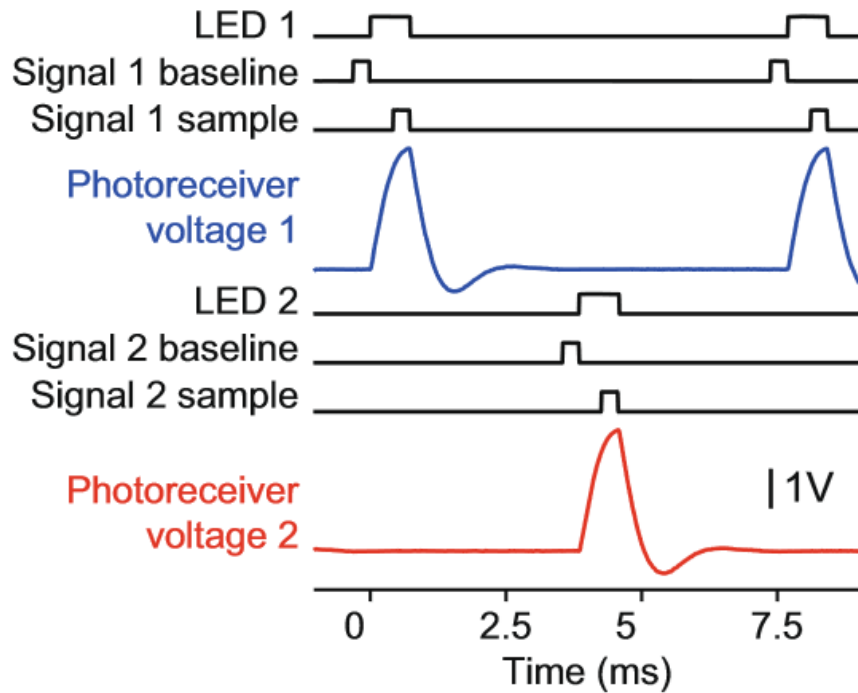
pyPhotometry GUI

- Acquisition from multiple setups
- Visualisation of signals and digital inputs
- Save and load experiment configurations
- Data saved either in custom binary format or .csv file, acquisition parameters saved with signals
- Implemented in Python + PyQt

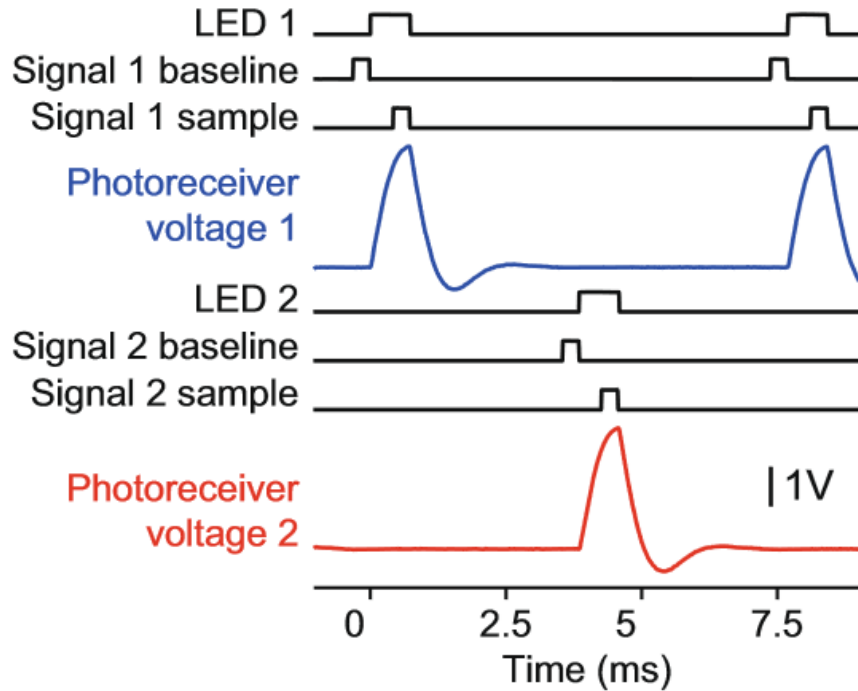


Time-division illumination

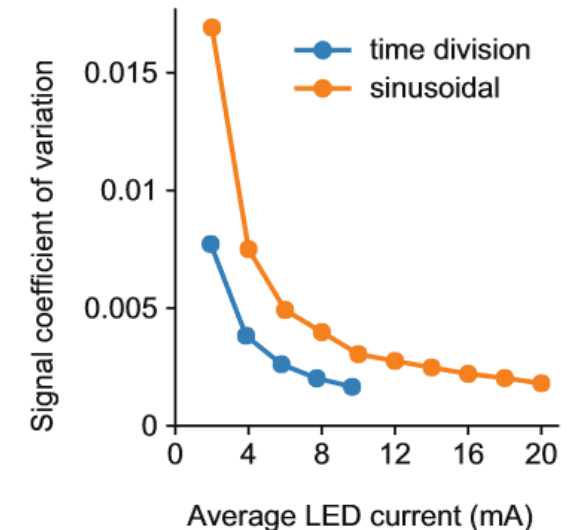
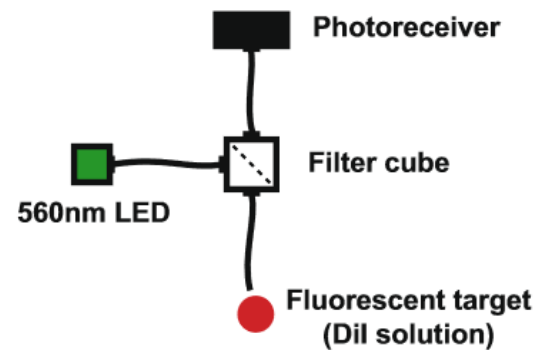
- Alternately pulsing LEDs allows selective readout of fluorescence evoked by different excitation wavelengths.
- Baseline subtraction removes influence of background illumination and low frequency noise.



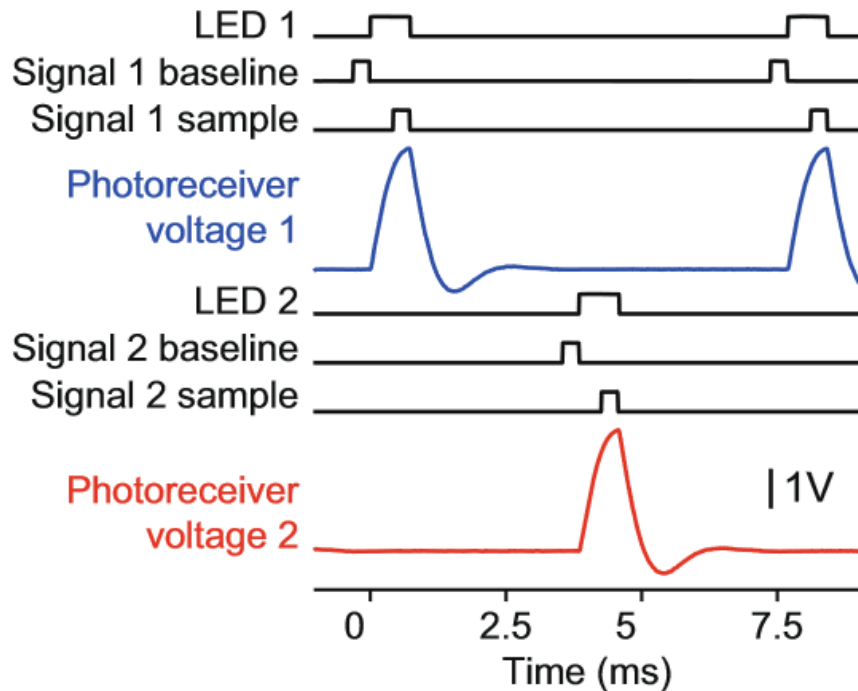
Time-division illumination



- Alternately pulsing LEDs allows selective readout of fluorescence evoked by different excitation wavelengths.
- Baseline subtraction removes influence of background illumination and low frequency noise.
- Time-division illumination appears to offer better signal-to-noise than frequency-division (sinusoidal) modulation schemes.



Time-division illumination



- Alternately pulsing LEDs allows selective readout of fluorescence evoked by different excitation wavelengths.
- Baseline subtraction removes influence of background illumination and low frequency noise.
- Time-division illumination appears to offer better signal-to-noise than frequency-division (sinusoidal) modulation schemes.
- Supported acquisition modes:
 - 2 Excitation λ 1 Emission λ (e.g. green + isosbestic)
 - 2 Excitation λ 2 Emission λ (e.g. red + green)
 - 3 Excitation λ 2 Emission λ (e.g. red + green + isosbestic)*

*with external LED driver.

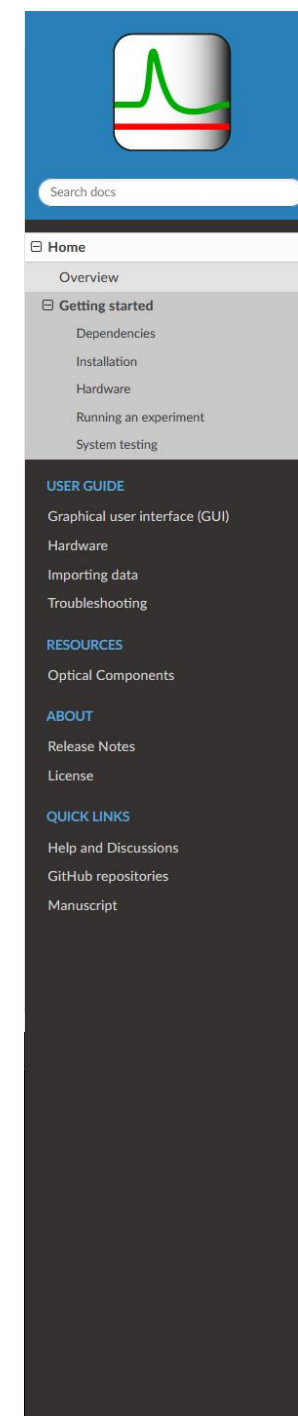
Documentation & support

- Detailed online documentation:

<https://pyphotometry.readthedocs.io>

- Github discussions board for user support:

<https://github.com/orgs/pyPhotometry/discussions>



»Home

Edit on GitHub

pyPhotometry

Open source, Python based, fiber photometry data acquisition.



Overview

pyPhotometry is system of open source hardware and software for neuroscience fiber photometry data acquisition, consisting of an [acquisition board](#) and [graphical user interface](#).

pyPhotometry supports data acquisition from two analog and two digital inputs, and control of two LEDs via built in LED drivers with an adjustable 0-100mA output. The system supports time-division multiplexed illumination which allows fluorescence evoked by different excitation wavelengths to be independently readout from a single photoreceiver signal. Assembled acquisition boards are available from the [OpenEphys store](#)

pyPhotometry hardware, software and documentation repositories are hosted on [GitHub](#), along with a [discussion board](#) for user support and questions.

The pyPhotometry [manuscript](#) has validation experiments characterising system performance, and information about the multiplexed illumination acquisition modes. If you use pyPhotometry in a publication, please cite it.

If your experiments involve behavioural tasks, you may be interested in pyPhotometry's sister project [pyControl](#), a Python based system for controlling behavioural experiments.

Getting started

Dependencies

pyPhotometry has the following dependencies:

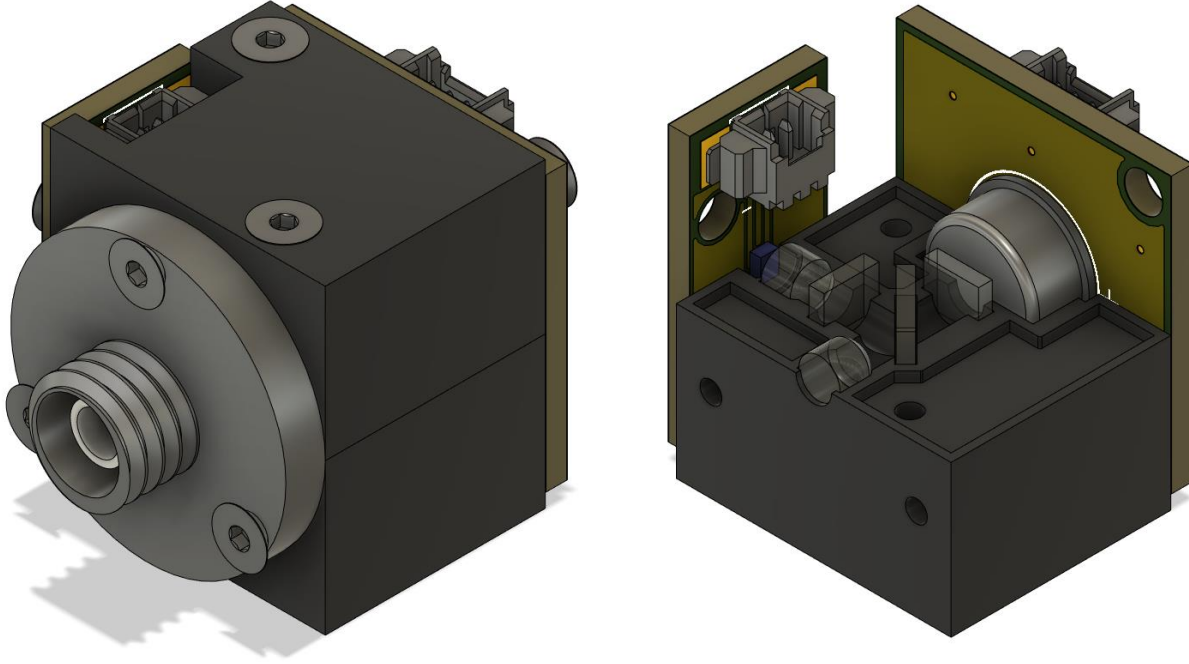
Optics setup



- Doric minicube with green (470nm Ex. 520nm Em.) and red channels (560nm Ex. 630nm Em.)
- 200um 0.5NA patch cords and optic fiber cannulae (RWD).
- Optionally Doric pigtailed rotary joint.

Future directions – Open source optics and photoreceiver

Mini-cube



Photoreceiver PCB

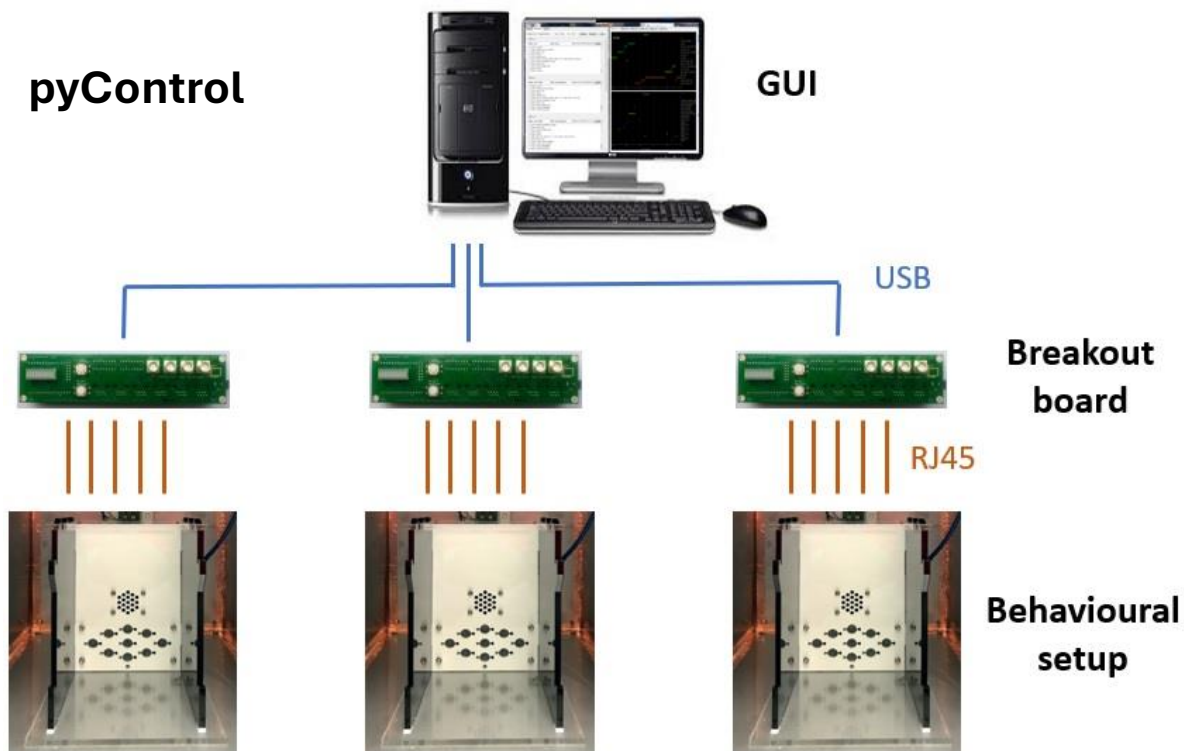


- 2×10^{10} V/A amplification
- 0 - 600Hz bandwidth
- 7mV output noise SD

pyControl & pyMultivideo

pyControl

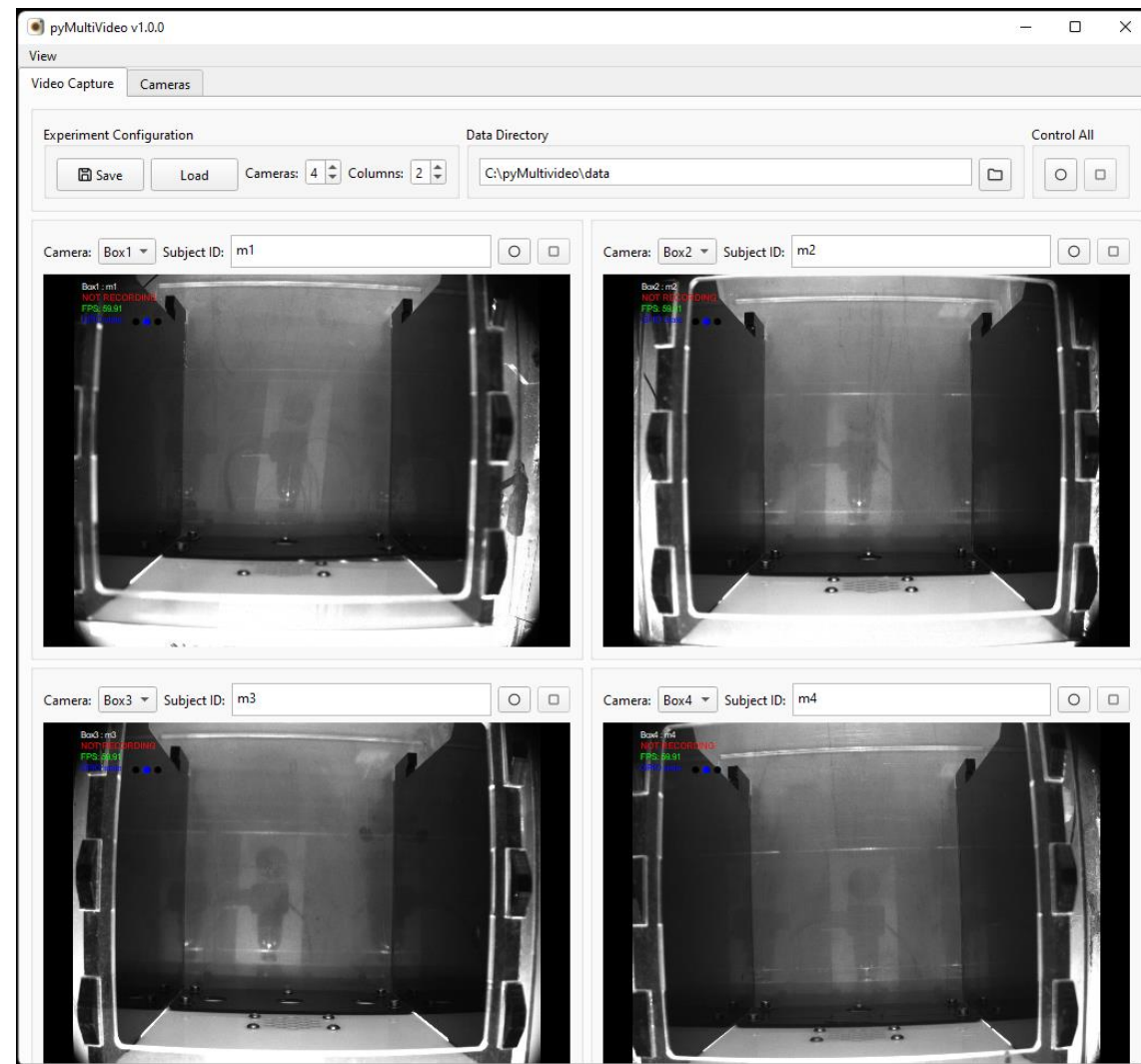
GUI



Behavioural experiment control

<https://pycontrol.readthedocs.io/>

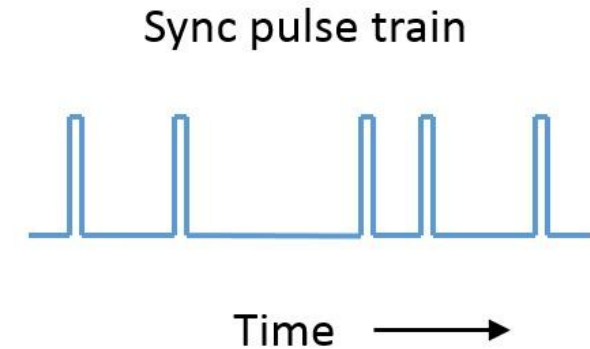
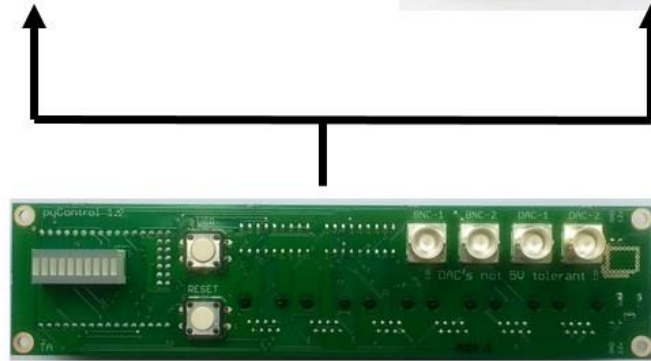
pyMultivideo



Video acquisition, synchronisation and compression

<https://pymultivideo.readthedocs.io>

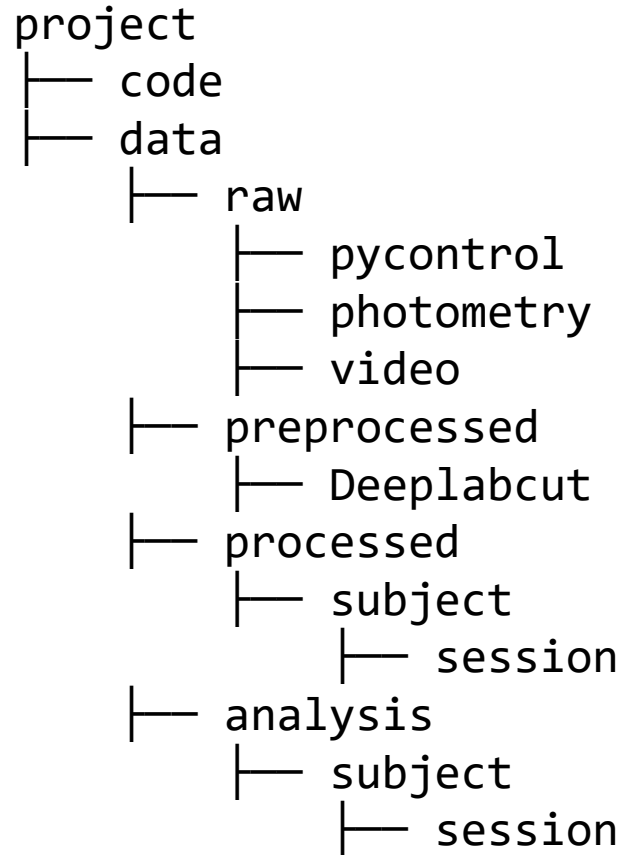
Synchronisation



- Output sync pulses with random inter pulse intervals (IPIs) from behavioural hardware (pyControl).
- Sync pulses are recorded by other systems in their own reference frame (e.g. sample / frame number)
- Random IPIs ensure unique match between sync pulse trains.
- Python module for automatically converting times between systems using sync pulse times.

<https://pycontrol.readthedocs.io/en/latest/user-guide/synchronisation>

Data organisation



- Raw data:
 - Files generated by acquisition hardware.
 - Organised by data type.
- Preprocessed data:
 - Output of packages like DeepLabCut or Kilosort.
 - Organised by data type.
- Processed data:
 - Version of the data used for analysis and data sharing.
 - Simple, generic filetypes (tsv, json, npy), based on IBL.
 - Times synchronised across data types.
 - Organised by subject – session.
 - Minimal redundancy.
- Analysis data:
 - Intermediate data structures generated during analysis.
 - Organised by subject – session.

Photometry data preprocessing

Aim: Remove variation from the photometry signal that is not due to the physiological process of interest.

Detected light:

- Sensor fluorescence
- Fiber & brain autofluorescence
- Ambient light

Signal variation:

- Sensor binding
- Photobleaching
- Noise
- Movement
- Physiological processes (blood flow, PH)

Relative contribution will depend on excitation and emission wavelengths

470nm Excitation 520nm Emission

Autofluorescence	Sensor fluorescence
------------------	---------------------

405nm Excitation 520nm Emission

Autofluorescence	Sensor fluorescence
------------------	---------------------

560nm Excitation 630nm Emission

Autofluorescence	Sensor fluorescence
------------------	---------------------

Photometry data preprocessing

Raw signals



Denoising



Photobleaching correction

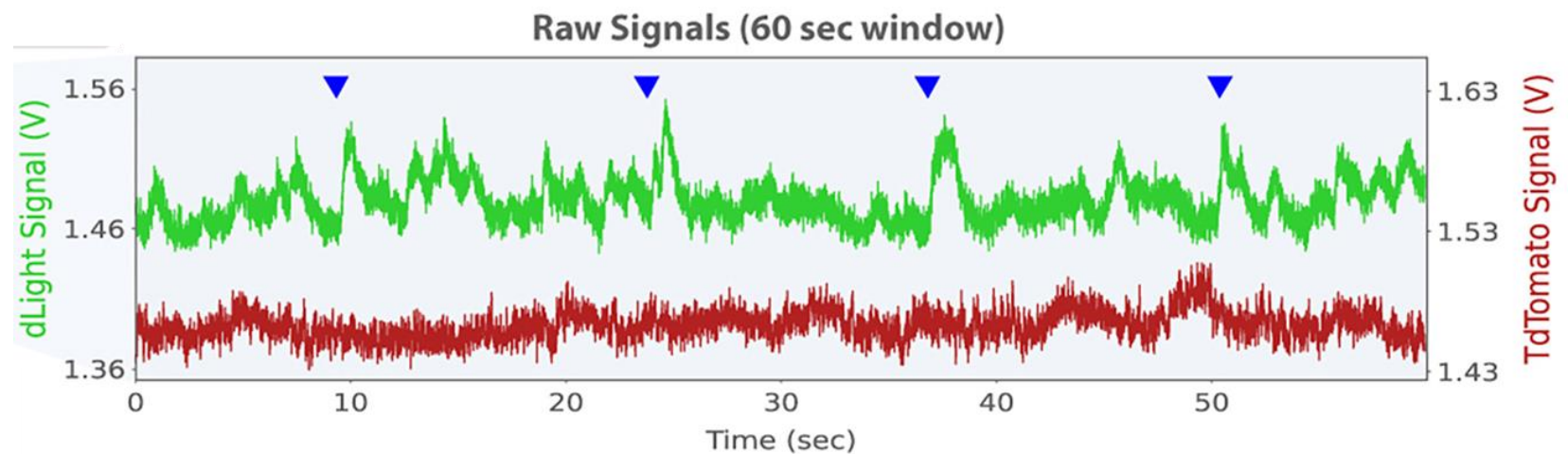
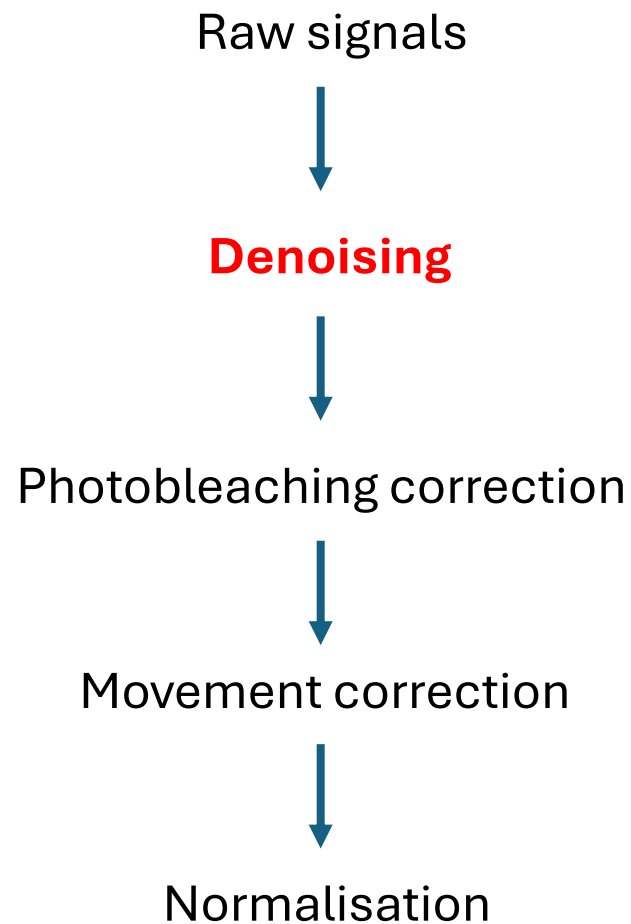


Movement correction

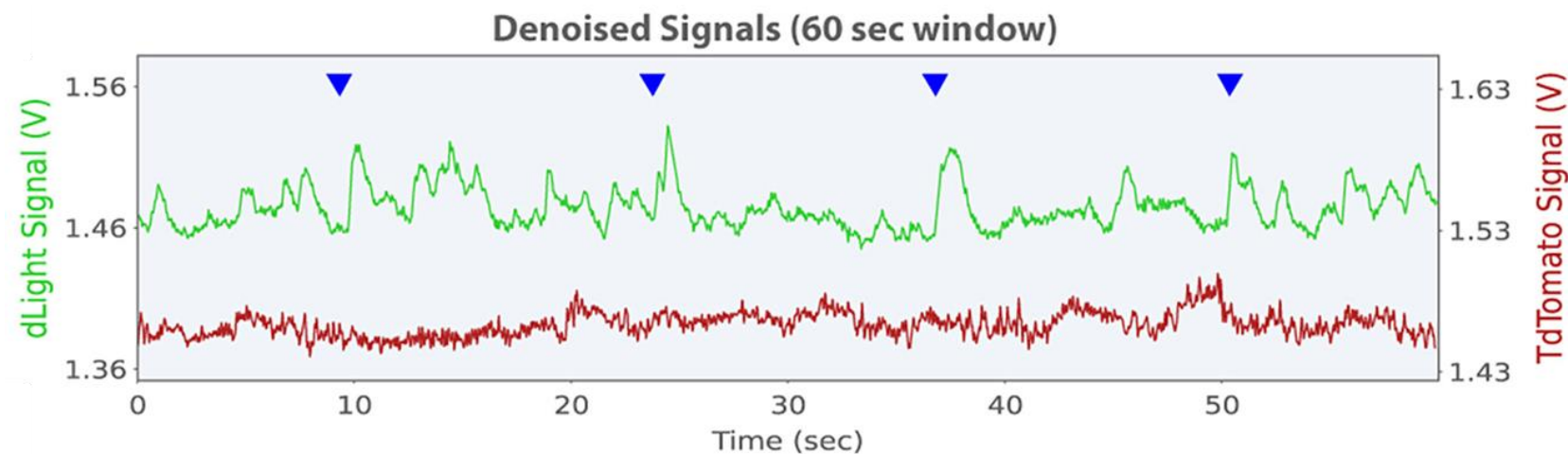


Normalisation

Photometry data preprocessing



↓ 10 Hz lowpass filter



Photometry data preprocessing

Raw signals



Denoising



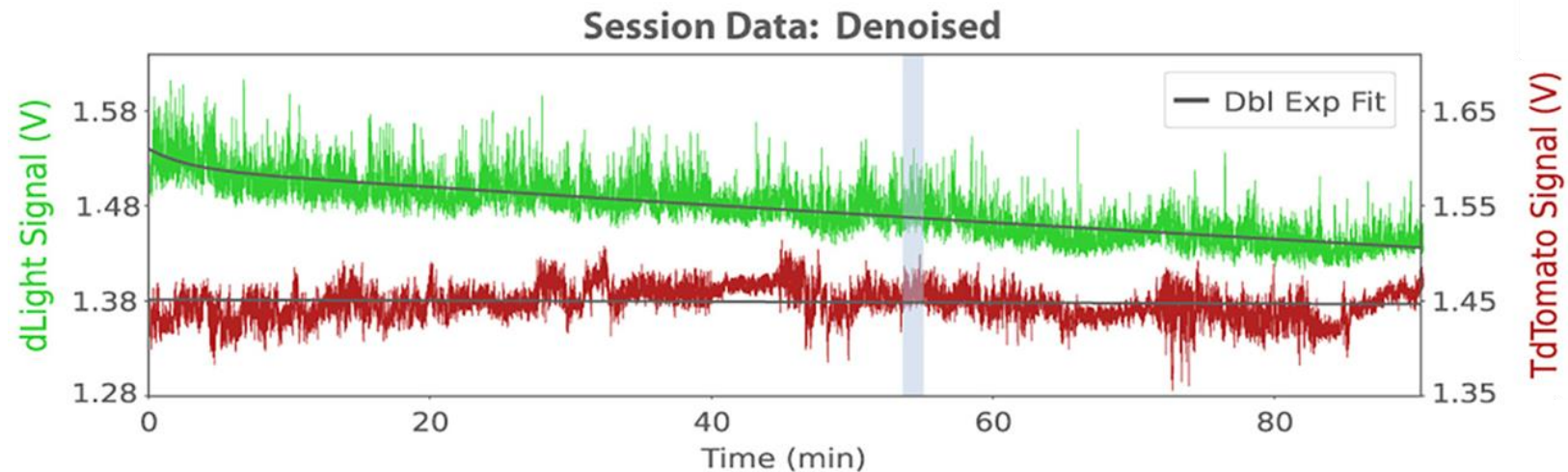
Photobleaching correction



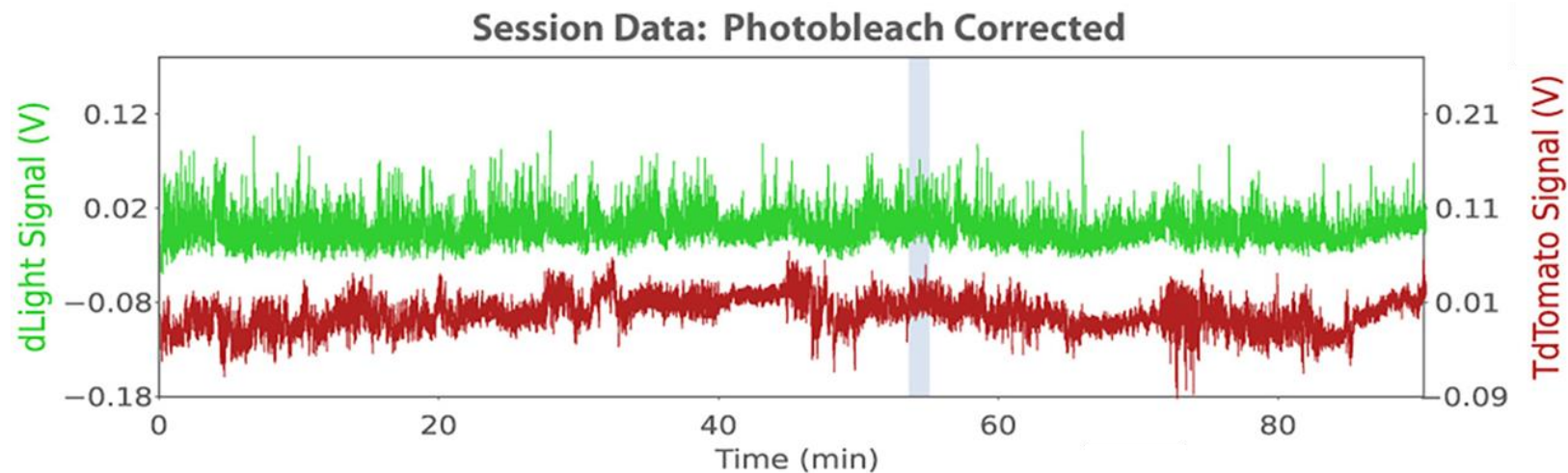
Movement correction



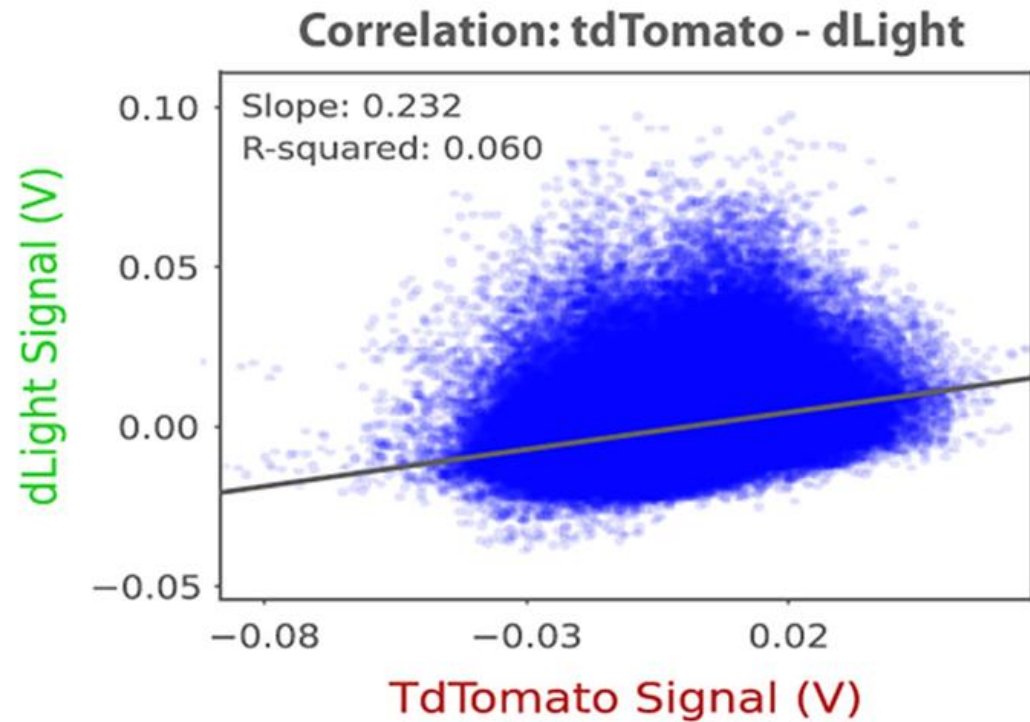
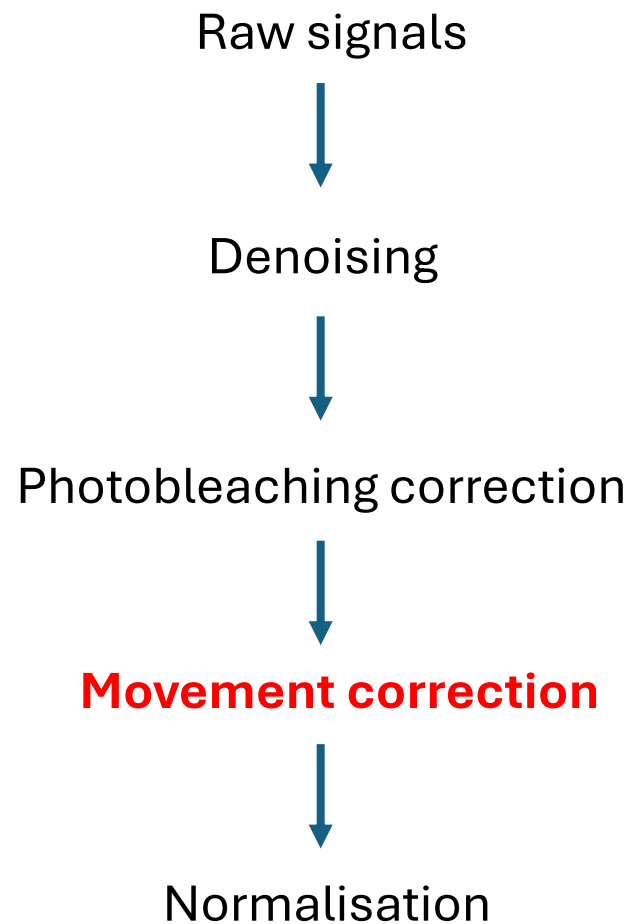
Normalisation



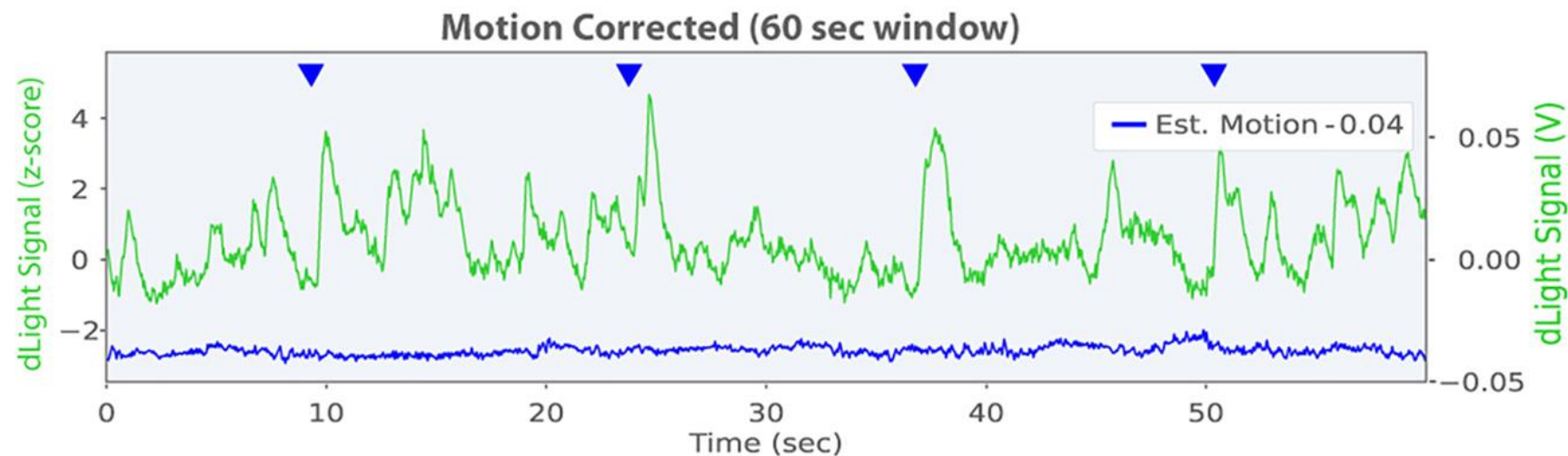
Subtract double exponential fit



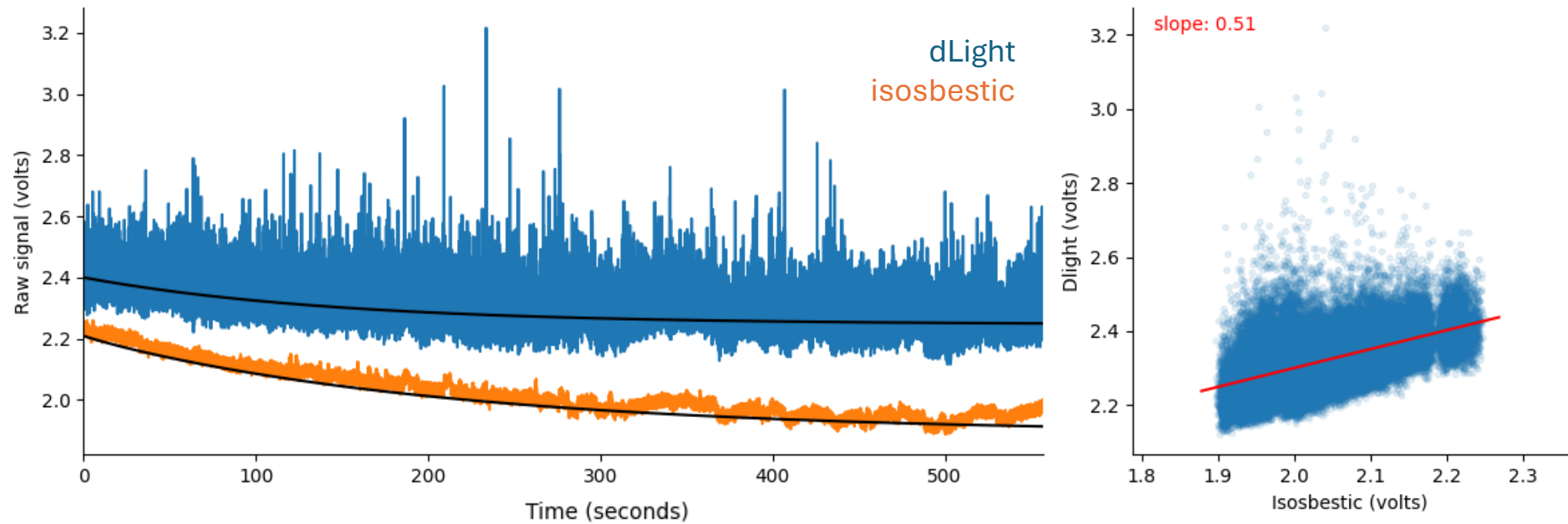
Photometry data preprocessing



- Estimate motion artifacts as linear fit of control channel to signal

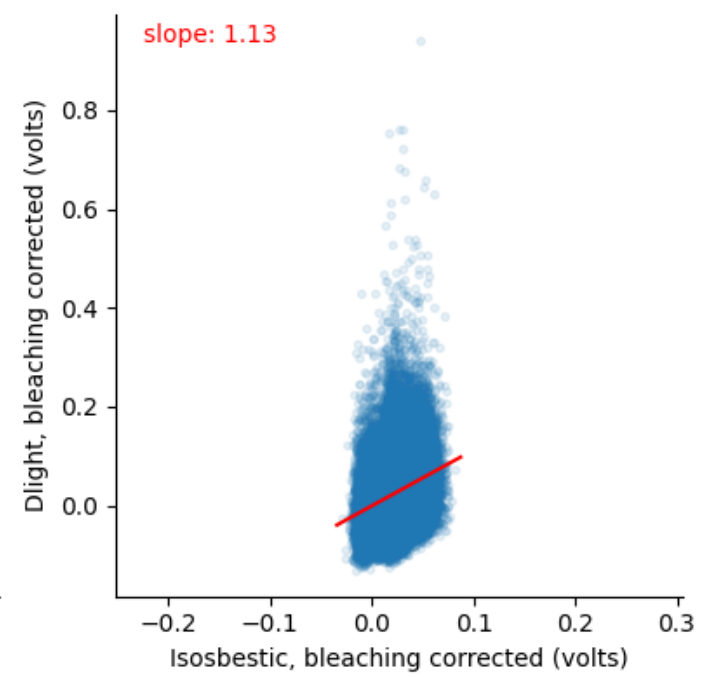
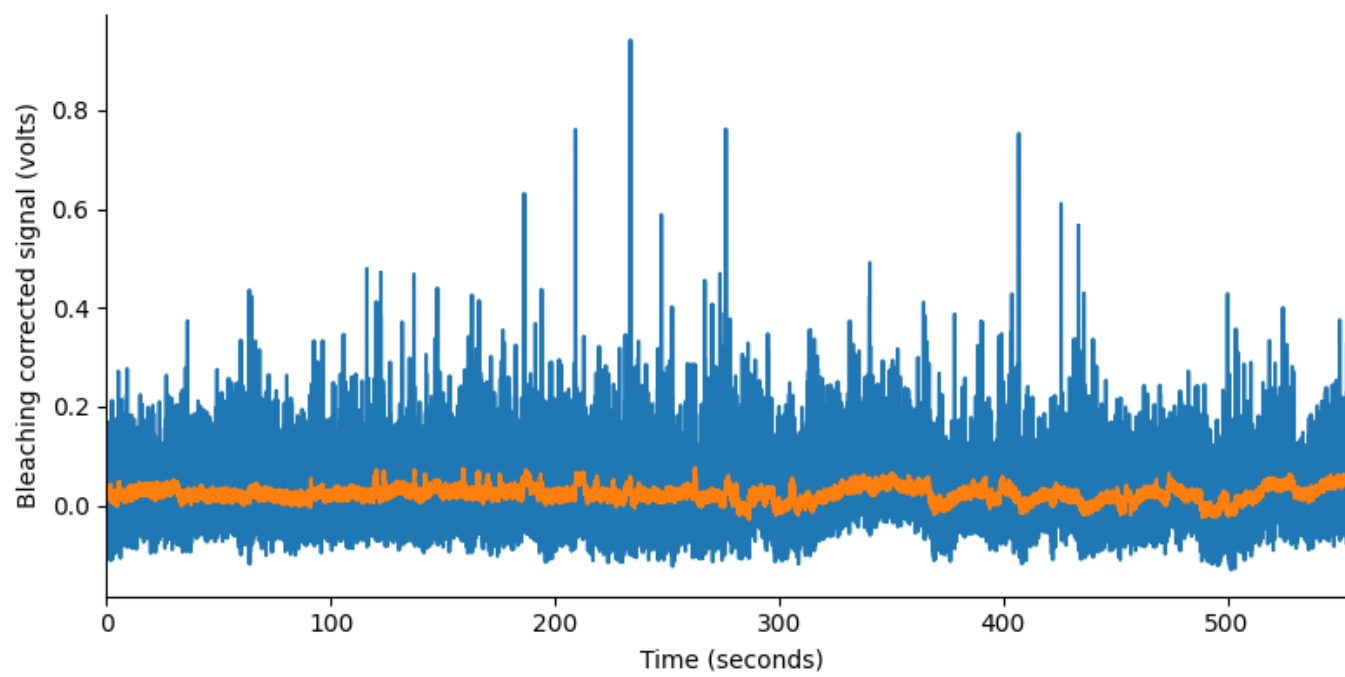
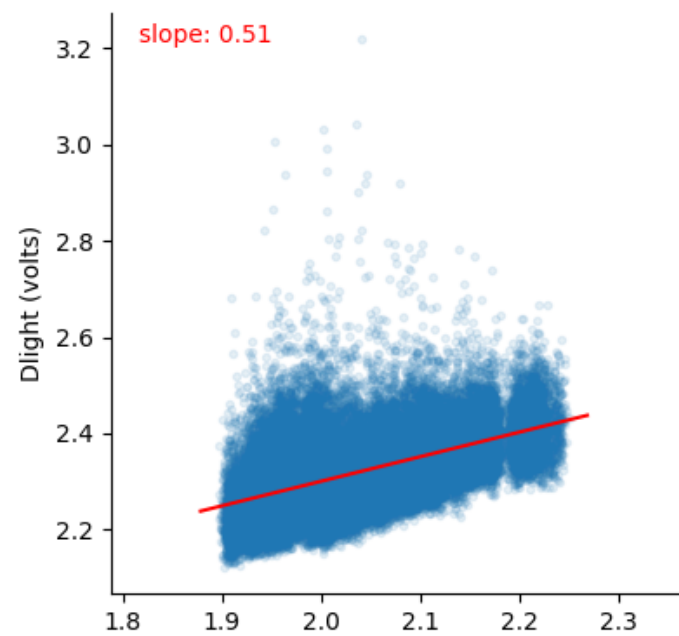
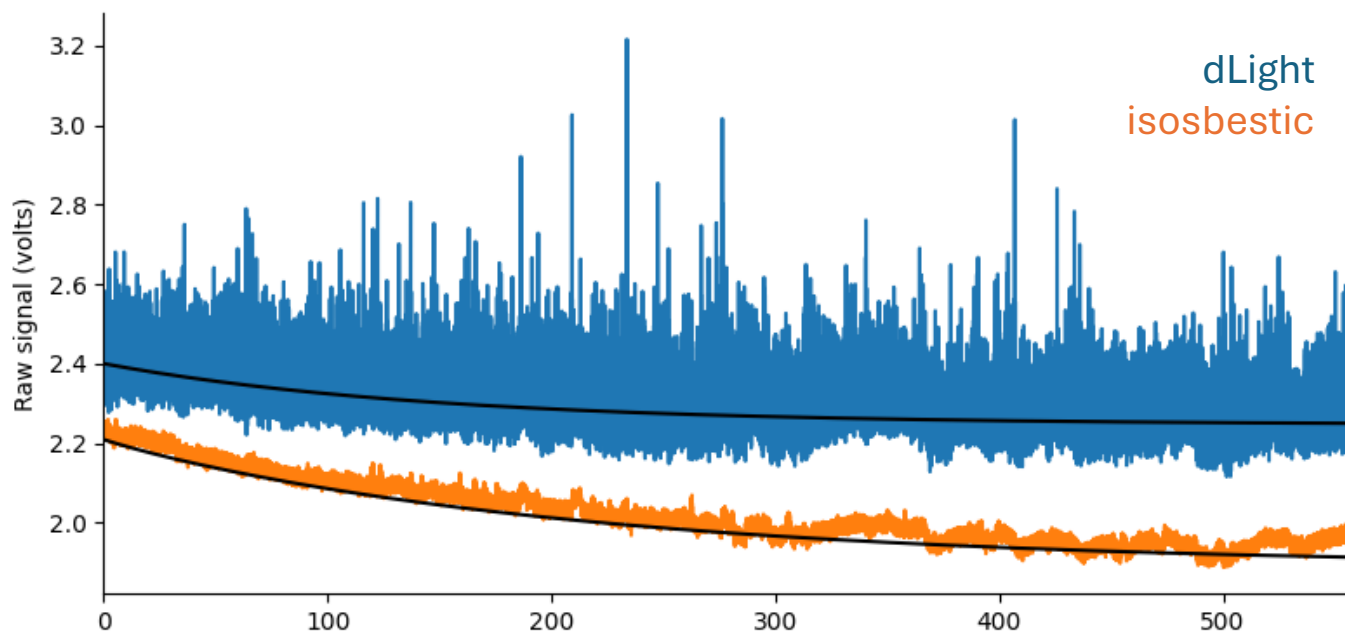


Combining photobleaching and movement correction

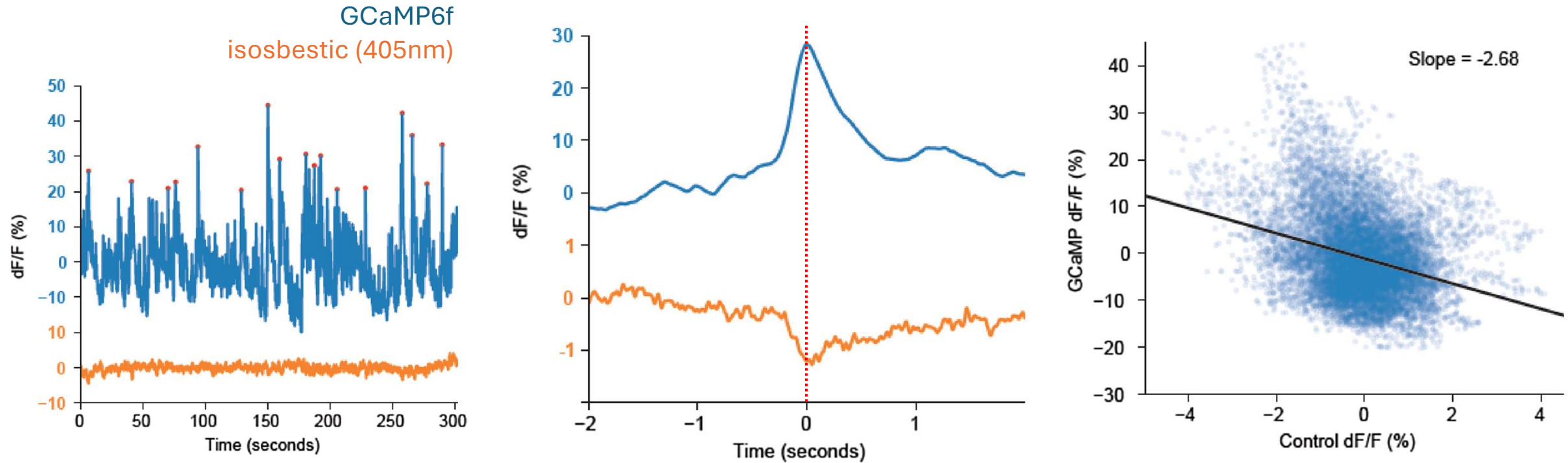


- Photobleaching and movement correction are often combined when using an isosbestic movement control channel, by linearly fitting the control channel to signal then dividing to give dF/F .
- Is this a good idea given that fluorophore and autofluorescence likely contribute different to the two channels?

Combining photobleaching and movement correction



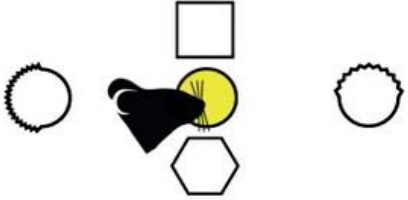
More movement control considerations



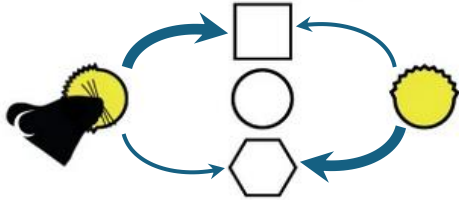
- If excitation is not exactly at isosbestic wavelength, some fluorescence change will occur with sensor binding.
- This can cause positive or negative bleed-through of signal into control channel.
- If signal bleed-through dominates control channel variance movement artifact estimation is not possible.

Photometry data analysis

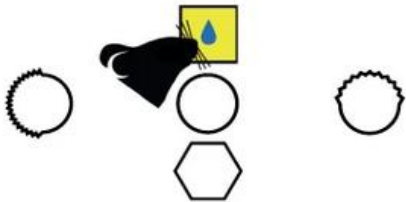
1. Initiate trial in center port



2. Choose between left and right ports



3. Poke active up/down port for probabilistic reward

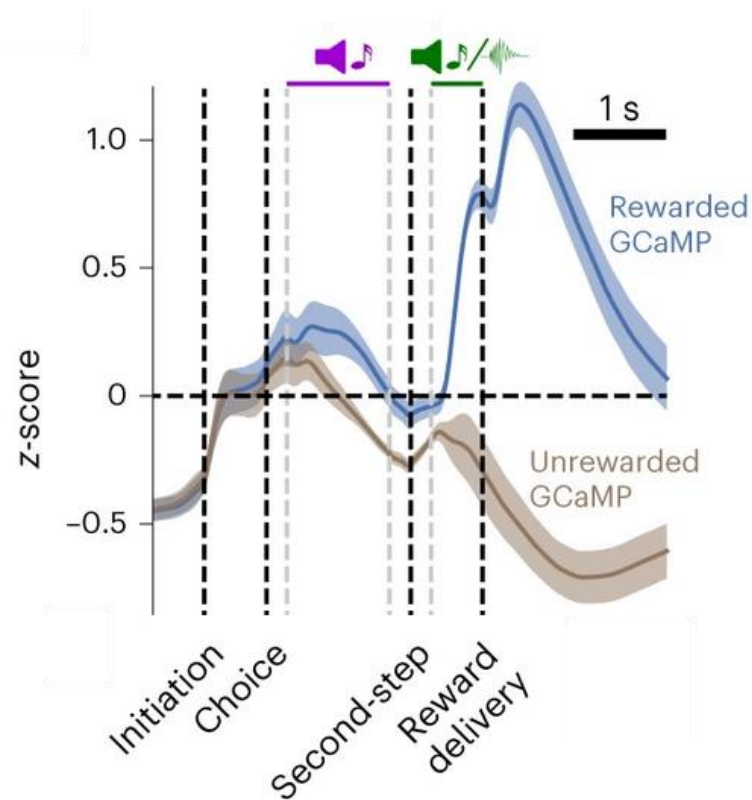


pre-processed signal

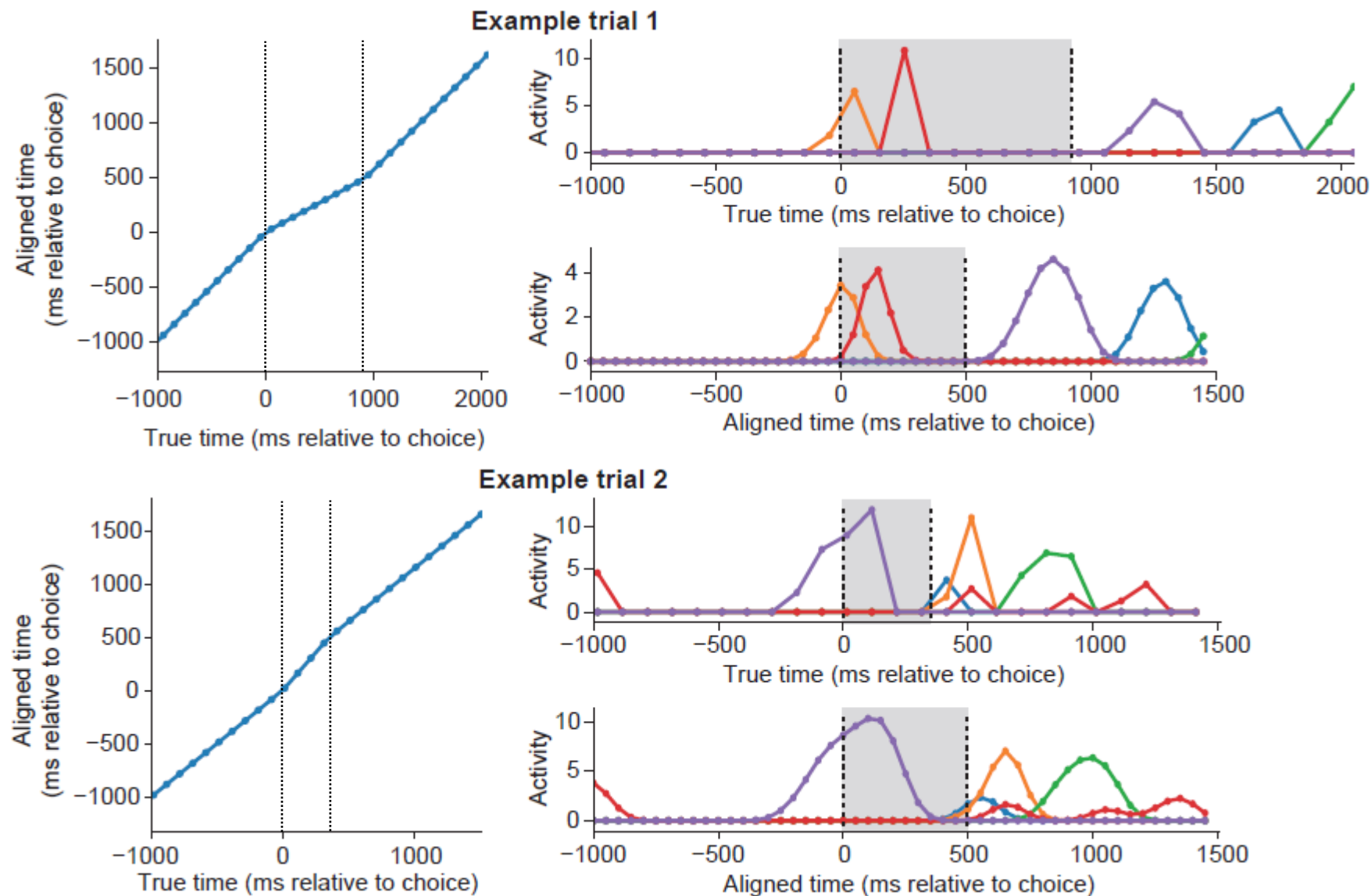
event or trial aligned signal

Plot responses

Linear regression

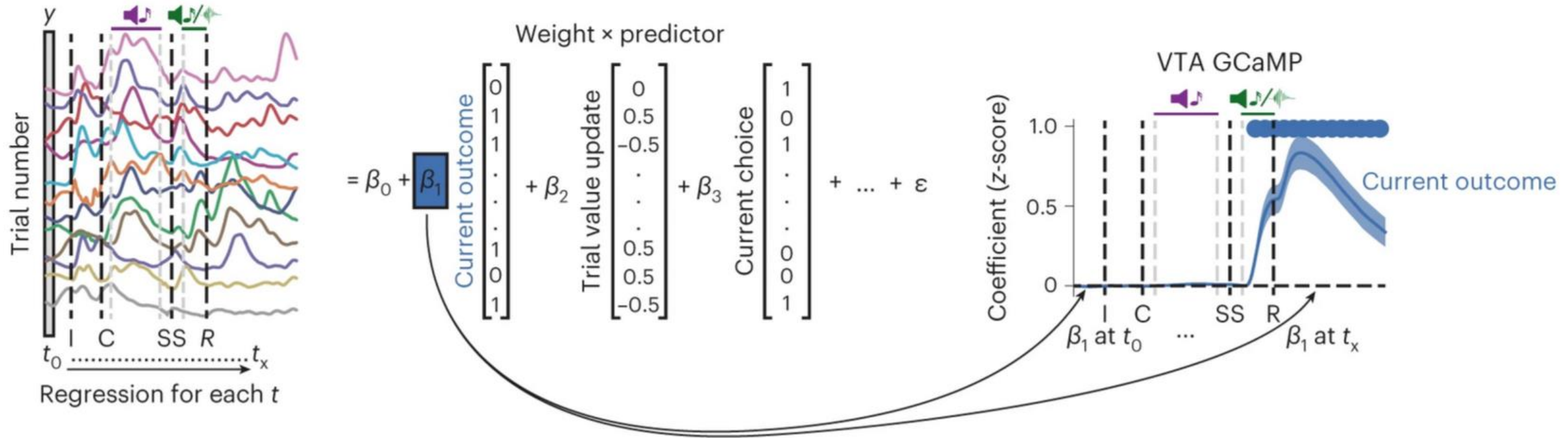


Trial aligning signals



- Align activity across trials by warping signals to match inter-event durations to median trial timings.
- Warp signal by linearly transforming sample times then Gaussian smoothing around output sample timepoints.

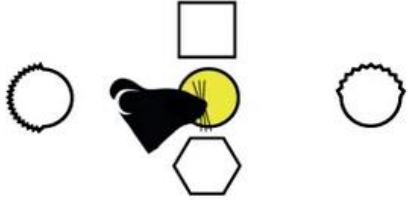
Trial-aligned linear regressions



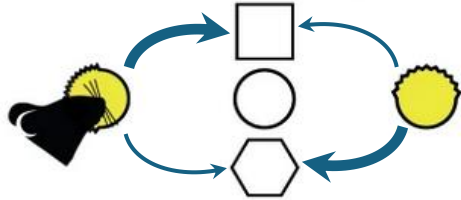
- Separate regression analyses for each timepoint in trial-aligned data, for each subject.
- Predictors values change from trial-to-trial, typically behavioural variables (outcomes, choices, values etc).
- Time course of weights (β 's) for each predictor indicate when and how each predictor explains signal variation.
- Statistical significance assessed with t-test of cross-subject distribution of β 's against 0, with false discovery rate multiple comparison correction for multiple timepoints.

Trial-aligned linear regressions

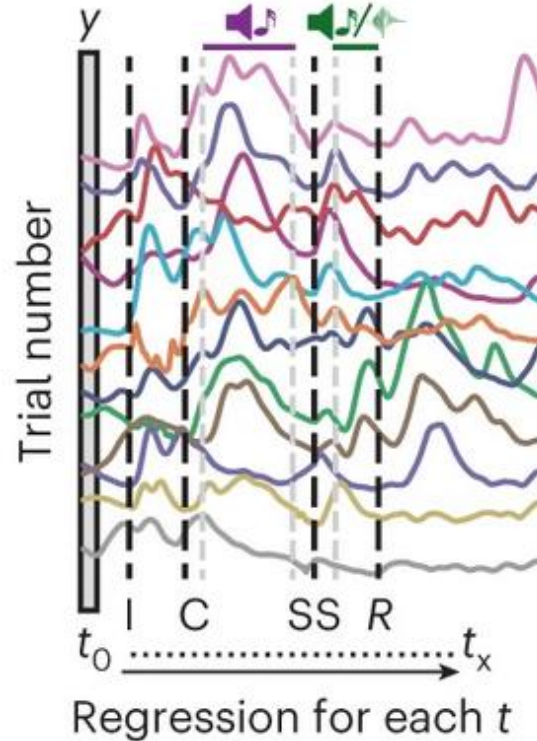
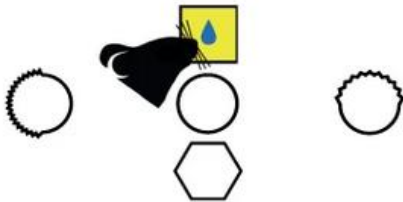
1. Initiate trial in center port



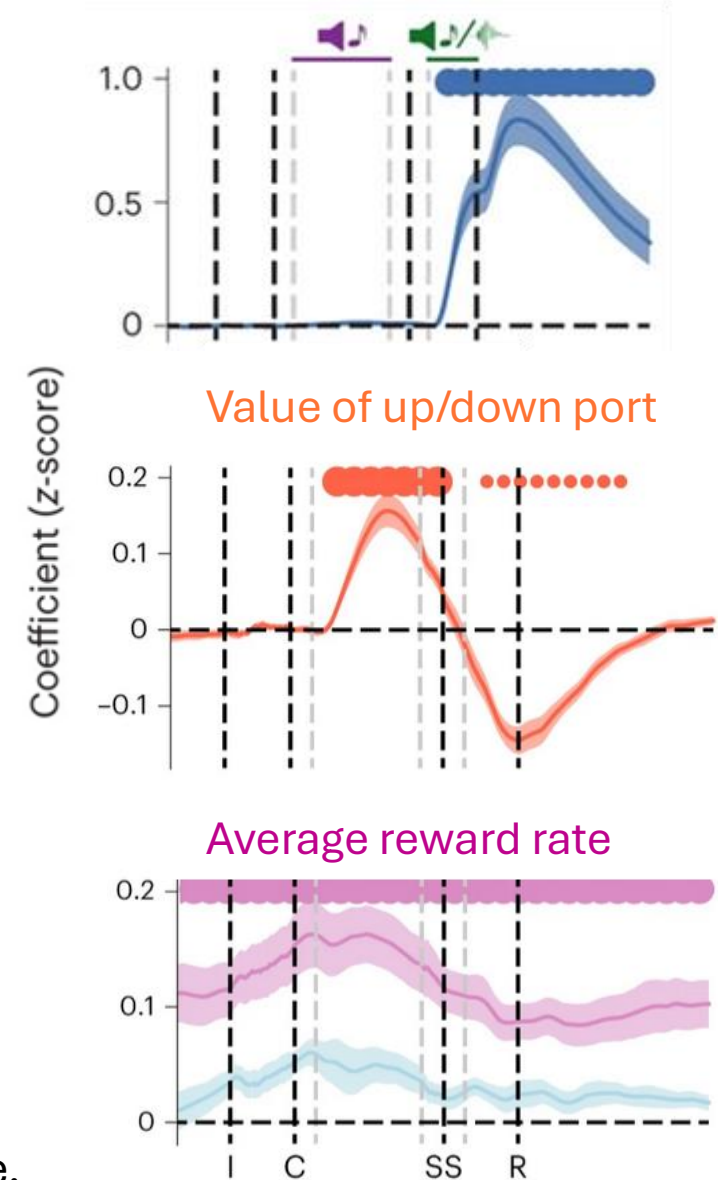
2. Choose between left and right ports



3. Poke active up/down port for probabilistic reward



Trial outcome (rewarded or not)



- State values and average reward rate explain separate variance in the NAc dopamine.

Summary:

Open source tools:

<https://pyphotometry.readthedocs.io>

<https://pycontrol.readthedocs.io/>

<https://pymultivideo.readthedocs.io>

Tutorial notebooks:

https://github.com/thomasAkam/photometry_preprocessing

https://github.com/cognitive-circuits/photometry_analysis

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